



R.K.D.F. UNIVERSITY, RANCHI

M.Sc. PHYSICS

SCHEME

S.N .	Subject Code	Subject Name	Marks Distribution				
			Internal	External		Total	
			Max	Max	Min	Max	Min
1.	PPH101	Mathematical Methods in Physics	30	70	21	100	35
2.	PPH102	Classical Mechanics	30	70	21	100	35
3.	PPH103	Quantum Mechanics-I	30	70	21	100	35
4.	PPH104	Classical Electrodynamics	30	70	21	100	35
<i>Practical</i>							
Total							
			Max		Min		
1.	PPH155	Practical-I -Lab	100		50		



R.K.D.F. UNIVERSITY, RANCHI

M.Sc. PHYSICS

**M.Sc. Physics 2019-2021
Choice Based Credit System**

S.N.	Subject Code	Subject Name	L	T	P	Credit
1.	PPH101	Mathematical Methods in Physics	3	1	0	4
2.	PPH102	Classical Mechanics	3	1	0	4
3.	PPH103	Quantum Mechanics-I	3	1	0	4
4.	PPH104	Classical Electrodynamics	3	1	0	4
5.	PPH155	Practical-I –Lab	0	0	8	4
Total			12	4	8	20



R.K.D.F. UNIVERSITY, RANCHI

M.Sc. PHYSICS

Syllabus SEMESTER – I

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc (Physics)	MATHEMATICAL METHODS IN PHYSICS	PPH101

Unit - I

Complex Analysis : Complex numbers, Complex functions (polynomials, Exponential, Trigonometric complex functions, Logarithm). Limits and continuity, differentiation, Analytical functions, Cauchy-Riemann conditions, Rectifiable arcs, Line integrals, Cauchy's theorem, Cauchy integral formula, Derivatives of analytical functions, Liouville's theorem. Power series Taylor's theorem, Laurent's theorem. Calculus of residues, evaluation of real definite integrals, summation of series, elementary discussion of branch cuts, Applications : Principal value integrals and dispersion relations.

Unit - II

Fourier series, Fourier integrals, Fourier transform, Parseval Relations, Convolution, Applications, Laplace transform, Bromwich integral (without proof) simple applications. Power series and contour integral solutions of second order diff. equations (Legendre, Bessel, Hermite, Laguerre as special examples properties of these functions). Legendre polynomials, Spherical harmonics and associated Legendre polynomials. Hermite polynomials. Sturm-Liouville systems and orthogonal polynomials.

Unit - III

Linear spaces and operators : Vector spaces, Linear independence, Bases, dimensionality isomorphisms. Linear transformations, inverses, matrices, similarity transformations, Eigenvalues and Eigenvectors. Inner product, orthogonality and completeness, complete orthogonal set, Gramm Schmidt orthogonalization procedure, Self- adjoint and unitary transformations. Eigenvalues and Eigenvectors of Hermitian and Unitary transformations, diagonalization. Function space and Hilbert space. Complete orthonormal sets of functions. Weierstrass's theorem (without proof) approximation by polynomial.

Unit - IV

Coordinate transformation in N-dimesional space: Contravariant and covariant tensor, Jacobian. Relative tensor, pseudo tensors (Example: change density, angular momentum) Algebra of tensors, Metric theorem, Associated tensors, Reimannian space (Example: Euclidian space and 4-D Minkowski space), Christoffel symbols, transformation of Christoffel symbols, Covariant differentiation. Ricci's theorem, Divergence, Curl



R.K.D.F. UNIVERSITY, RANCHI

M.Sc. PHYSICS

and Laplacian in tensor form. Stress-and Strain tensors. Hook's law in tensor form. Lorentz Covariance of Maxwell equation.

Unit - V

Group of transformations. (Example: symmetry transformation of square), Generators of a finite group, Normal subgroup, Direct product of groups.. Isomorphism and Homomorphism. Representation theory of finite groups, Invariant subspace and reducible representations, irreducible representation, Crystallo-graphic point groups. Irreducible representation of C_{4v} Translation group and the reciprocal lattice.

Reference Books:

1. Mathematical Methods for Physicists: George Arkfen (AcademicPress). .
2. Applied Mathematics for Engineers and Physicists: L. A. Pipe (McGraw Hill)
3. MathematicalMethods-Potter and Goldberg (Prentice Hall of fudia).
4. Elements of Group Theory for Physicists: A. W. Joshi (Wiley Eastern Ltd.)
5. VectorAnalysis (Schaum Series) (Mc Graw Hill).



R.K.D.F. UNIVERSITY, RANCHI

M.Sc. PHYSICS

Syllabus SEMESTER – I

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc (Physics)	CLASSICAL MECHANICS	PPH102

UNIT-I

Holonomic and nonholonomic constraints: D'Alembert's Principle, Generalized coordinates, Lagrangian, Lagrange's equation and its applications, Velocity dependent potential in Lagrangian formulation. Generalized momentum, Legendre transformation, Hamiltonian, Hamilton's Canonical equation. Calculus of variations and its application to simple problems, Hamilton's variational principle, Derivation of Lagrange's and Hamilton's Canonical equation from Hamilton's variational principle. Extension of Hamilton's Principle for nonconservative and nonholonomic systems. Method of Lagrange's multipliers,

Unit - II

Conservation principle and Noether's theorem. Conservation of energy, linear momentum and angular momentum as a consequence of homogeneity of time and space and isotropy of space respectively.

Unit-III

Canonical transformation, integral invariants of Poincaré: Lagrange's and Poisson brackets as canonical invariants. cyclic coordinates, Equation of motion in Poisson bracket formulation, Infinitesimal contact transformation and generators of symmetry, Liouville's theorem, Hamilton Jacobi equation and its applications.

Unit - IV

Action angle, variable adiabatic invariance of action variable : The Kepler problem in action angle variables, theory of small oscillation in Lagrangian formulation, normal coordinates and its applications, Orthogonal transformation, Eulerian angles, Euler theorem, Eigen values of the inertia tensor, Euler equations. Force free motion of a rigid body.

Unit - V

Laplace transforms, and their properties, Laplace transform of derivatives and integrals of Laplace transform, Laplace, Convolution theorem, Impulsive function Application of Laplace transform in solving linear differential equations with constant coefficient with variable coefficient and linear partial differential equation.

Reference Books:

- 1 Goldstein Classical Mechanics.
2. Landau and Lifshitz - Classical Mechanics.
3. A. Raychoudhary - Classical Mechanics.



R.K.D.F. UNIVERSITY, RANCHI

M.Sc. PHYSICS

Syllabus SEMESTER – I

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc(Physics)	CLASSICAL ELECTRODYNAMICS	PPH104

Unit-I

1. **Electrostatics: Electric field;** Gauss law, Differential form of Gauss law. Another equation of electrostatics and the scalar potential, surface distribution of charges and dipoles and discontinuities in the electric field and potential, Poisson and Laplace equations, Green's Theorem, Uniqueness of the solution with Dirichlet or Neumann Boundary conditions, Formal solution of Electrostatic Boundary value problem with Green's Function, Electrostatic potential energy and energy density, capacitance. **Boundary- Value Problems in Electrostatics:** Methods of Images, Point charge in the presence of a grounded conducting sphere point charge in the presence of a charge insulated conducting sphere, Point charge near a conducting sphere at fixed potential, conducting sphere in a uniform electric field by method of images, Green function for the sphere, General solution for the potential, Conducting sphere with Hemispheres at different potential, orthogonal functions and expansion.

Unit-II

2. **Magnetostatics:** Introduction and definition, Biot, and Savart law, the differential equation of magnetostatics and Ampere's law, Vector potential and Magnetic induction for a circular current loop, Magnetic fields of a localized current distribution, Magnetic moment, Force and torque on and energy of a localized current distribution in an external magnetic induction, Macroscopic equations. Boundary conditions on B and H. Methods of solving Boundary-value problems in magnetostatics, Uniformly magnetized sphere, Magnetized sphere in an external field, Permanent magnets, Magnetic shielding, spherical shell of permeable material in a uniform field.

Unit - III

Multipoles, Electrostatics of Macroscopic Media Dielectrics: Multiple expansion, multipole expansion of the energy of a charge distribution in an external field, Elementary treatment of electrostatics with permeable media, Boundary value problems with dielectrics. Molar polarizability, and electric susceptibility. Models for molecular polarizability, Electrostatic energy in dielectric media. .

Time varying fields, Maxwell's equations Conservation Laws: Energy in a magnetic field, Vector and Scalar potentials. Gauge transformations, Lorentz gauge, Coulomb gauge, Green functions for the wave equation, Derivation of the equations of Macroscopic Electromagnetism, Poynting's theorem and conservations of energy and momentum for a system of charged particles. and EM fields. Conservation laws for macroscopic media. Electromagnetic field tensor. Transformation of four potentials and four currents. Tensor description of Maxwell's equation.



R.K.D.F. UNIVERSITY, RANCHI

M.Sc. PHYSICS

Unit - IV

Plane Electromagnetic Waves and Wave Equation : Plane wave in a nonconducting medium. Frequency dispersion characteristics of dielectrics, conductors and plasmas, waves in a conducting or dissipative medium, superposition of waves in one dimension, group velocity, causality connection between D and E. Kramers-Kronig relation.

Covariant Form of Electrodynamical Equations : Mathematical properties of the space-time special relativity, Invariance of electric charge covariance of electrodynamics, Transformation of electromagnetic fields.

Unit - V

Radiation by moving charges : Lienard-wiechert Potentials for a point charge, Total power radiated by an accelerated charge, Larmor's formula and its relativistic generalization, Angular distribution of radiation emitted by an accelerated charge, Radiation emitted by a charge in arbitrary extremely relativistic motion. Distribution in frequency and angle of energy radiated by accelerated charges, Thomson scattering and radiation, Scattering by quasi free charges, coherent and incoherent scattering, Cherenkov radiation.

Reference Books :

1. J.D. Jackson--Classical Electrodynamics
2. Panofsky and Philips Classical Electricity and Magnetism'
3. Introduction to Electrodynamics-Griffiths
4. Landau and Lifshitz--Classical Theory of Field
5. Landau and Lifshitz.:Electrodynamics of Continuous Media



R.K.D.F. UNIVERSITY, RANCHI

M.Sc. PHYSICS

Syllabus SEMESTER – I

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc (Physics)	QUANTUM MECHANICS-I	PPH103

UNIT - I

States, Amplitudes and Operators: States of a quantum mechanical system, representation of quantum, - mechanical states, properties. of quantum mechanical amplitude; operators and change of state, a complete set of basis states, Products of linear operators, language of quantum mechanics, postulates, essential definitions and commutation relations.

Observable and description of system: Process of measurement, expectation values, time dependence of quantum mechanical amplitude, observables with no classical analogue, spin. dependence of quantum-mechanical amplitude on position. the wave function, super-position of amplitudes, identical particles.

UNIT – II

The Co-ordinate Representation: Compatible observables, quantum conditions and uncertainty relation, Co- ordinate representation, of operator & position, momentum and angular momentum, time dependence of expectation values, the Ehrenfest's theorem; the time evolution of wave function, the Schrodinger equation, energy quantization, ,periodic potential as an example.

Symmetries and Angular momentum: (a) Compatible observables and constants of motion, symmetry' transformation and conservation laws, invariance, under space and time translations and space rotation and conservations of momentum, energy and angular momentum.

UNIT - III

Angular momentum operators and, their eigen values, matrix representations of the angular momentum operators and their eigen states, co-ordinate representations of the orbital angular momentum operators and their eigen state (spherical harmonics), composition of angular momentum, Clebsch- Gordon coefficients tensor operators and Wigner Eckart theorem, commutation relations, of J_x , J_y , J_z with reduced spherical tensor operator, matrix elements of vector operators, time reversal invariance and vanishing of static electric dipole moment of a stationary state.

UNIT-IV

Hamiltonian matrix and the time evolution of Quantum mechanical States: Hermiticity of the Hamiltonian matrix, Time independent perturbation of an arbitrary system, simple matrix examples of time-, independent perturbation, energy given states of a two state system, diagonalizing of energy matrix, time independent perturbation of two state system the perturbative solution: Weak field and strong field cases, general description of two state system. Pauli matrices. Ammonia molecule as an example of two state system.



R.K.D.F. UNIVERSITY, RANCHI

M.Sc. PHYSICS

UNIT - V

Interaction with External Fields: Non degenerate first order stationary perturbation method, atom in a weak uniform external electric field and first and second order Stark effect, calculation of the polarizability of the ground state of H-atom and of an isotropic harmonic oscillator, Degenerate stationary perturbation theory. Linear Stark effect for H-atom levels, inclusion of spin-orbit and weak magnetic, field, Zeeman effect, strong magnetic field and calculation of interaction energy.



R.K.D.F. UNIVERSITY, RANCHI

M.Sc. PHYSICS

Second Semester

S.N.	Subject Code	Subject Name	Marks Distribution				
			Internal	External		Total	
			Max	Max	Min	Max	Min
1.	PPH201	Spectroscopy	30	70	21	100	35
2.	PPH202	Nuclear Physics – I	30	70	21	100	35
3.	PPH203	Quantum Mechanics –II	30	70	21	100	35
4.	PPH204	Electronics Digital Circuits	30	70	21	100	35
<i>Practical</i>							
Total							
			Max		Min		
1.	PPH255	Practical-II Lab	100		50		



R.K.D.F. UNIVERSITY, RANCHI

M.Sc. PHYSICS **M.Sc. Physics 2019-202 1** **Choice Based Credit System**

S.N.	Subject Code	Subject Name	L	T	P	Credit
1.	PPH201	Spectroscopy	3	1	0	4
2.	PPH202	Nuclear Physics – I	3	1	0	4
3.	PPH203	Quantum Mechanics –II	3	1	0	4
4.	PPH204	Electronics Digital Circuits	3	1	0	4
5.	PPH255	Practical-II Lab	0	0	8	4
Total			12	4	8	20



R.K.D.F. UNIVERSITY, RANCHI

M.Sc. PHYSICS

Syllabus SEMESTER – II

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc (Physics)	Spectroscopy	PPH201

UNIT- I

Atomic Spectra: Space quantization, Relation between angular momentum and magnetic moment, Bohr magneton. Fine structure of spectral lines, Term symbols of alkali and alkaline earth atoms. LS and JJ coupling. Quantum theory of Zeeman effect (normal and anomalous), Paschen-Back effect, Stark effect (linear and non-linear). Hyperfine structure of spectral lines, X-ray spectra characteristics and absorption.

UNIT- II

The Rotation of the Molecule: Rotational spectra-Rigid diatomic molecule, The intensities of spectral lines, Effect of isotopic substitution, the non-rigid rotator, Simple harmonic oscillator, The anharmonic oscillator, Diatomic vibrating rotator, Born Oppenheimer approximation, Techniques and instrumentation applications.

UNIT- III

Molecular Spectra: Infrared and Raman spectra of diatomic molecules using an-harmonic oscillator, non-rigid rotator and vibrating rotator as models. Electronic states and electronic transitions in diatomic molecules, Frank Condon principle.

UNIT- IV

Resonance Spectroscopy: Nature of spinning particle, Interaction between spin and a magnetic field, Larmor Precession, Theory of NMR, Chemical shift-relaxation Mechanism, experimental study of NMR, Theory and experimental study of NQR, Theory of ESR, Hyperfine structure and fine structure of ESR, Experimental studies and applications, Mossbauer spectroscopy, Principle-Isomer shift, Quadrupole effect, effect of magnetic field, Instrumentation applications.

UNIT- V

Laser and Holography: Spontaneous and stimulated emission, Einstein A and B coefficients, Basic Principles of Laser, Population Inversion-Two level and Three level Laser system, optical pumping³⁺ rate equation, modes of resonator and coherence length, The Nd, YAG laser, The Neodymium Glass laser, The CO₂ Laser, Organic Dye lasers, Semi-conductor Laser, Liquid Laser. Principle of Holography, Theory-practical applications including data storage.

Books Suggested:

- Kuhn, "Atomic Spectra".
- Ghatak & Loknathan, "Quantum Mechanics".
- Herzberg, Spectra of diatomic molecules
- Elements of Spectroscopy: Gupta, Kumar and Sharma, PragatiPrakashan.
- Fundamentals of Molecular Spectroscopy: Colin and Elaine, TMH.
- Laser and Non-linear Optics: B.B.Laud, New Age Publications.



R.K.D.F. UNIVERSITY, RANCHI

M.Sc. PHYSICS

Syllabus SEMESTER – II

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc (Physics)	ELECTRONICS DIGITAL CIRCUITS	PPH204

UNIT- I

Operational Amplifiers: .Differential amplifier - circuit configurations-dual input, balanced output differential amplifier. DC analysis - AC analysis, inverting and noninverting inputs, CMRR - constant current bias level translator. Block diagram of a typical Op-Amp-analysis.Open loop configuration,inverting and non-inverting amplifiers

UNIT- II

Op-amp with negative feedback - voltage series feed back -effect of feed back on closed loop gain, input resistance, output resistance, bandwidth and output offset voltage - voltage follower. Practical op-amp-input offset voltage -input bias current -input offset current, total output offset voltage, CMRR frequency response. DC and AC amplifier, summing, scaling and averaging amplifiers, instrumentation amplifier, integrator and differentiator.

UNIT - III

Oscillators and Wave Shaping Circuits: Oscillator Principle- Oscillator types, Frequency stability, response, The Phase shift oscillator, Wein bridge Oscillator,LC tunable oscillators, Multivibrators-Monostable and Astable, Comparators, Square wave and Triangle wave generation, Clamping and Clipping. Voltage regulators- fixed regulators, Adjustable voltage regulators, Switching regulators.

UNIT - IV

Digital Electronics: Combinational Logic :The transistor as a switch;circuit Realisation of OR,AND,NOT, NOR and NAND gates, Exclusive OR gate, Boolean algebra - Demorgan's theorems Adder, Subtractor, Comparator, Decoder / Demultiplexer ,Data selector/ multiplexer -Encoder.

UNIT-V

Sequential Logic: Flip -Flops: one-bit memory; The RS Flipflop, JK Flip- Flop, JK master slave Flip -Flops, T Flip - Flop, D Flip- Flop, Shift registers - synchronous and asynchronous counters- cascade counters, Binary counter, Decade counter.

Basic concepts about fabrication and characteristics of integrated circuits. Fortran 77: Variable, Expression, jumping. Branching an looping statement ,Input / Output statement for handling Input / Output Files, Subroutine, External, Function ,Special statements ,COMMON,ENTRY FORMAT,PAUSE, Equivalence . Programming of simple problems involving use of interpolation differentiation, Integration, matrix inversion and least square analysis.



R.K.D.F. UNIVERSITY, RANCHI

M.Sc. PHYSICS

Reference Book

1. Ryder-Electronic Fundamentals and applications.
2. Millman and Thub-Pulse, Digital and Switching waveforms.
3. Millman and Helkias-Integrated Electronics.
4. Ryder-network Lines and Fields.
- 5 Bapat-Electronics Devices and Circuitrs.
6. A Ralston and P. Rabinowitz, A First Course in Numerical analysis Mc Graw Hill (1985)
7. S.S. Sastry, Introductory Methods of Numerical Analysis. Prentice hall of India (1979).
8. Ram Kumar, Programming with Fortran 77, McGraw-Hill (1986).
9. "Electronic'Devicesand circuit theory by Robert Boylested and Louis Nashdsky PHI, New Delhi. 1100001, 1991 .
10. "OPAmps& Linear integratedcircuits, by Ramakanth A. Gayakwad PHI, Second Edition, 1991.
11. Digital principles and Applications by A.P. Malvino and Donald P.Leach, Tata Megraw - Hill company, New Delhi, 1993.
12. Microprocessor Architecture, Programming and applications with 8085/8086 by Ramesh S. Gaonkar,Wiley - Eastern Ltd., 1987.



R.K.D.F. UNIVERSITY, RANCHI

M.Sc. PHYSICS

Syllabus SEMESTER – II

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc(Physics)	NUCLEAR PHYSICS-I	PPH202

UNIT - I

Nucleon-Nucleon Scattering and Potentials : Partial wave analysis of the neutron-proton scattering at low energy assuming central potential with square well shape, concept of the-scattering length, coherent scattering of neutrons by protons in (ortho and para) hydrogen molecule; conclusions of these analyses regarding scattering lengths, range and depth of the potential; the effective range theory (in neutron-proton scattering) and the shape independence of nuclear potential; A qualitative discussion of proton proton scattering at low energy: General features of two-body scattering at high energy Effect of exchange forces: Phenomemonological Hamada- Johnston hard core potential and Reid hard core and soft core potentials; Main features of the One boson Exchange Potentials (OBEP) no derivation.

UNIT - II

Two Nucleon system and Nuclear Forces: General nature of the force between nucleons, saturation of nuclear forces, charge independence and spin dependence, General forms of two nucleon interaction, central, non central and velocity dependent potentials, Analysis of the ground state (3S1)of deuteron using a square well potential, range-depth relationship, excited states of deuteron, Discussion of the ground state of deuteron under non central force, calculation of the electric quadrupole and magnetic dipole moments and the D-state admixture.

Experimental Techniques: Gas filled counters; Scintillator counter, Cerenkov counters; Solid state detectors; Surface barrier detectors; Electronic circuits used with typical nuclear detectors; Multiwire proportion chambers; Nuclear emulsions, techniques of measurement and analysis of tracks; Proton synchrotron; Linear accelerations; Acceleration of heavy ions.

UNIT - III

Nuclear shell model: Single particle and collective motions in nuclei: Assumptions and justification of the shell model, average shell potential, spin orbit coupling; single particle wave functions and level sequence; magic numbers; shell model predictions for ground state parity; angular momentum, magnetic dipole and electric- quadrupole moments; and their comparison with experimental data; configuration mixing; single particle transition probability according to the shell model; selection rules; approximate estimates for the transition probability and Weisskopf units: Nuclear isomerism.

Collective nuclear models: Collective variable to describe the the cooperative modes of nuclear motion; Parametrization of nuclear surface; A brief description of the collective model Hamiltonian (in the quadratic approximation); Vibrational modes of a spherical nucleus, Collective modes of a deformed even-even nucleus and moments of, inertia; Collective spectra and electromagnetic transition in even nuclei and comparison with experimental data; Nilsson model for the single particle states in deformed nuclei.



R.K.D.F. UNIVERSITY, RANCHI

M.Sc. PHYSICS

UNIT - IV

Interaction of radiation and charged particle with matter (No derivation): Law of absorption and attenuation coefficient; Photoelectric effect, Compton scattering, pair production; Klem-Nishima cross sections for polarized and unpolarized radiation, angular distribution of scattered photon and electrons, Energy loss of charged particles due to ionization, Bremstrahlung; energy target and projectile dependence of all three processes, Range-energy curves; Straggling.

Nuclear Reactions: Theories of Nuclear Reactions; Partial wave analysis of reaction Cross section; Compound nucleus formation and breakup, Resonance scattering and reaction- Breit-Wigner dispersion formula for S-waves ($l=0$), continuum cross section; statistical theory of nuclear reactions, evaporation probability and cross section for specific reactions; The optical model, Stripping and pick-up reactions and their simple theoretical description (Butler theory) using plane wave Born approximation (PWBA) Short comings of PWBA nuclear structure studies with deuteron stripping (d,p) reactions.

UNIT - V

Nuclear gamma and beta decay: Electric and magnetic multipole moments and gamma decay probabilities in nuclear system (no derivations), Reduced transition probability, General characteristics of weak interaction; nuclear beta decay and lepton capture; electron energy spectrum and Fermi- Kurie plot; Fermi theory of beta decay (parity conserved selection rules Fermi and Gamow-Teller) for allowed transitions; f -values; General interaction Hamiltonian for beta decay with parity conserving and non conserving terms; Forbidden transitions
,Experimental verification of parity violation; The V-A interaction and experimental evidence.

Reference Books :

1. J. M Blatt and V.E. Weisskopf: Theoretical Nuclear Physics
2. Statistical theory of nuclear reactions, Exaporation probability and cross section for specific reaction.
3. L.R.B Elton: Introductory Nuclear Theory, ELBS Pub. London, 1959
4. B.K. Agrawl : Nuclear Physics, Lokbharti Pub, Allahabad. 1989
5. M.K. Pal: Nuclear Structlire, Affiliated East-West Press, 1982).
6. RR Roy and B.P.Nigam, Nuclear Physics, Willey-Easter, 1979
7. M.A. Preston & RK Bhaduri-Structure of the Nucleus, Addison Wesley, 1975
8. RM. Singru : Introductory Experimental Nuclear Physics
9. England - Techniques on Nuclear Structure (Vol.D
10. RD. Evans-TheAtomicNucleus(McGraw-Hills, 1955)
11. H. Enge -Introduction to Nuclear PeYsic~,Addition-Wesley, 1970
12. W.E.Burcham- Elementsof NuclearPhysics,ELBS, Longman, 1988
13. B.L. Cohen - Concept of Nuclear Physics Tata Mc-Graw Hills, 1988
14. E. Segre - Nuclei, Particles Benjamin, 1977
15. I. Kaplan - Nuclear Physics, Addison Wesley, 1963
16. D. Hallidy - Introductory Nuclear Physics, Wiley, 1955.
17. Harvey - Introduction of Nuclear Physics and Chemistry



R.K.D.F. UNIVERSITY, RANCHI

M.Sc. PHYSICS

Syllabus SEMESTER – II

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc(Physics)	QUANTUM MECHANICS –II	PPH203

UNIT - I

Scattering (non-relativistic): Differential and total scattering cross section, - transformation from CM frame to Lab frame, solution of scattering problem by the method of partial wave analysis, expansion of a plane wave into a spherical wave and scattering amplitude, the optical theorem, Applications.- scattering from a delta potential, square well potential and the hard sphere scattering of identical particles, energy dependence and resonance scattering. Breit-Wigner formula, quasi stationary states. The Lippman-Schwinger equation and the Green's function approach for scattering problem, Born approximation and its validity for scattering problem, Coulomb scattering problem under first Born approximation in elastic scattering.

UNIT - II

Relativistic Formulation and Dirac Equation: Attempt for relativistic formulation of quantum theory, The Klein-Gordon equation, Probability density and probability current density, solution of free particle KG equation in momentum representation, interpretation of negative probability density and negative energy solutions.

Dirac equation for a free particle, properties of Dirac matrices and algebra of gamma matrices, non-relativistic correspondence of the Pauli equation (inclusive of electromagnetic interaction). Solution of the free particle. Dirac equation, orthogonality and completeness relations for Dirac spinors, interpretation of negative energy solution.

UNIT - III

Symmetries of Dirac Equation : Lorentz covariance of Dirac equation, proof of covariance and derivation of Lorentz boost and rotation matrices for Dirac spinors, Projection operators involving four momentum and spin, Parity (P), Charge conjugation(C), time reversal (T) and CPT operators for Dirac spinors, Bilinear covariants, and their transformations behavior under Lorentz transformation, P,C,T and CPT, expectation values of coordinate and velocity, involving only, positive energy solutions and the associated problems, inclusion of negative energy solution, Zitter bewegung, Klein paradox.

UNIT - IV

The Quantum Theory of Radiation : Classical radiation field, transversality condition, Fourier decomposition and radiation oscillators, Quantization of radiation oscillator, creation, annihilation and number operators; photon states, photon as a quantum mechanical excitations of the radiation field, fluctuations and the Uncertainty relation, validity of the classical description, matrix element for emission and absorption, spontaneous emission in the-dipole approximation, Rayleigh scattering. Thomson scattering and the -Raman effect, Radiation damping and Resonance fluorescence.



R.K.D.F. UNIVERSITY, RANCHI

M.Sc. PHYSICS

UNIT – V

Scalar and vector fields: Classical Lagrangian field theory, 'Euler-Lagrange's equation, Lagrangian density for electromagnetic field. Occupation number representation for simple harmonic oscillator, linear array of coupled oscillators, second quantization of identical bosons, second quantization of the real Klein Gordan field and complex, Klein-Gordon field, the meson propagator. The occupation number representation for fermions, second quantization of the Dirac field, the fermion propagator, the e.m. interaction and gauge invariance, covariant quantization of the free electromagnetic field, the photon propagator.

UNIT – VI

S-matrix, the S-matrix expansion, Wick's theorem, Diagrammatic representation in configuration space, the momentum representation, Feynman diagrams of basic processes, Feynman rules of QED. Applications of S.

Matrix formalism: the Coulomb scattering, Bhabha scattering, Moller scattering, Compton scattering and pair production.

Reference Books :

1. Ashok Das and A.C. Millisones : Quantum Mechanics -A Modern Approach.(Garden and Breach Science Publishers)
2. E. Merzbaker : Quantum Mechanics, Second Edition (John Wiley and sons)
3. Bjorken and Drell : Relativistic Quantum Mechanics (M.Graw Hill)
4. J.J. Sakurai : Advanced Quantum Mechanics (John Wiley)
5. F. Mandal & G. Shaw, Quantum Field Theory (John Wiley)
6. J.M. Ziman, Elements of Advance Quantum Theory, (Cambridge University Press.



R.K.D.F. UNIVERSITY, RANCHI

M.Sc. PHYSICS



R.K.D.F. UNIVERSITY, RANCHI

M.Sc. PHYSICS

PRACTICAL

I & II LIST OF EXPERIMENTS FOR M.Sc.

List of Experiments (any eighteen) :

1. To design a single stage amplifier of a given voltage gain and lower cut of frequencies.
2. To determine L , C , and R of a given coil and to study the variations of R with frequency.
3. To design a RC coupled two stage amplifier of a given gain and the cutoff frequencies.
4. To study Hartley oscillator.
5. To Study Transistor bias Stability.
6. To design a Multivibrator of given frequency and study its wave shape.
7. To study the characteristics of FET and use it to design a relaxation oscillator and measure its frequency.
8. To study the characteristics of an operational amplifier.
9. To study the characteristics of a UJT and use it to design a relaxation oscillator and measure its frequency.
10. To study the addition, integration and differentiation properties of an operational amplifier.
11. Determine Planck constant using solar Cell.
12. To determine Planck constant and work function by a photo-cell.
13. To study regulated power supply using (A) Zener diode only (b) Zener diode with a series transistor (c) Zener diode with a shunt transistor.
14. To verify Fresnel's formula;
15. To study the percentage regulation and variation of Ripple factor, with load for a full wave rectifier. .
16. To study analog to digital and digital to analog conversion.
17. To study a driven mechanical oscillator.
18. To verify Hartmann's formula using constant deviation spectrograph.
19. To find e/m of electron using Zeeman effect.
20. To find Dissociation energy to I.
21. Study of CH Bands.
22. Salt Analysis / Raman effect (Atomic).
23. Design and study of pass filters.
24. Michelson Interferometer.
25. Fabry parot Interferometer.
26. Determination of velocity of Ultrasonic waves.
27. Study of Elliptically polarised light by Babinet Compensator.
28. Verification of Cauchy's Dispersion relation.
29. Study of DC gate control characteristics and Anode current characteristics of SCR.



R.K.D.F. UNIVERSITY, RANCHI

M.Sc. PHYSICS

SCHEME

Semester – III

S.N.	Subject Code	Subject Name	Marks Distribution				
			Internal	External		Total	
			Max	Max	Min	Max	Min
1.	PPH301	Numerical Methods and Simulation	30	70	21	100	35
2.	PPH302	Statistical Physics	30	70	21	100	35
3.	PPH303	Nuclear Physics–II	30	70	21	100	35
4.	PPH304	Condensed Matter Physics –I	30	70	21	100	35
<i>Practical</i>							
Total							
			Max		Min		
1.	PPH355	Practical-III Lab	100		50		



R.K.D.F. UNIVERSITY, RANCHI

M.Sc. PHYSICS

Choice Based Credit System

S.N.	Subject Code	Subject Name	L	T	P	Credit
1.	PPH301	Numerical Methods and Simulation	3	1	0	4
2.	PPH302	Statistical Physics	3	1	0	4
3.	PPH303	Nuclear Physics–II	3	1	0	4
4.	PPH304	Condensed Matter Physics –I	3	1	0	4
5.	PPH355	Practical-III Lab	0	0	8	4
Total			12	4	8	20



R.K.D.F. UNIVERSITY, RANCHI

M.Sc. PHYSICS

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc (Physics)	NUMERICAL METHODS AND SIMULATION	PPH301

Unit-I

Solution of Linear System: Numerical solution of algebraic equation, Iteration, Newton Raphson method, Solution of Linear system, Direct method, Gauss, Gauss-Jordon elimination method, Matrix inversion and LU decomposition, Eigenvalues and Eigenvectors, Applications. Interpolation, Lagrange approximation, Newton and Chebyshev Polynomials, Least square fitting, Application in some physical problems.

Numerical Differentiation and Integration: Numerical solution of ordinary differential equation, Iteration method, Picard's method, Euler's method and improved Euler's method. Introduction to quadrature, trapezoidal and Simpson's rule Applications.

Numerical Solution of Partial Differential Equations: First and second order, Linear and non-linear differential equations, Solution by method of iteration, Euler and Runge Kutta methods. Finite difference method, Relaxation, Fourier and cyclic reduction and the Rayleigh-Ritz method, Application to diffusion of dopant in a semiconductor, Wave equation in a coaxial cable, Vibrating strings and membranes, Poisson equation, Schrodinger equations.

Monte Carlo Technique: Evaluation of single and multi-dimensional integrals, Optimization problems, Applications to statistical mechanics, Metropolis algorithm.

Simulation/ Modeling: Concept of modeling, Introduction to techniques of modeling, State variable model of system, Model parameters and simulation using SCILAB / MATLAB / SIMULINK, Time domain and frequency domain analysis of systems using SCILAB / MATLAB, Spice modeling of semiconductor devices (pn diode and BJT) and programming methodology, Circuit simulation using PSpice / LabView / ORCAD

Books Suggested:

- Introduction to Numerical Analysis, S.S. Sastry, PHI Learning Pvt. Ltd.
- Schaum's Outline of Programming with C++, J. Hubbard, McGraw-Hill Pub.
- Numerical Recipes in C: The Art of Scientific Computing W.H Press et al, Cambridge University Press.
- A First Course in Numerical Methods, U.M Ascher & C. Greif, PHI Learning.
- Elementary Numerical Analysis, K.E. Atkinson, Wiley India Edition



R.K.D.F. UNIVERSITY, RANCHI

M.Sc. PHYSICS

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc (Physics)	NUCLEAR PHYSICS –II	PPH303

Unit-I

Fundamental Properties of Nuclei: Electric moments and magnetic moments of nucleous, Measurement of magnetic moment of neutron, proton and nuclear magnetic moments, Parity and statics of nucleus, i-spin formalism.

Unit-II

Two-Nucleon Forces: Theory of ground state of the deuteron, Partial wave analysis of low energy n-p and p-p scattering, Effective range theory of low energy n-p and p-p scattering, Coherent n-p scattering and spin dependence of nuclear force, Exchange forces and tensor forces, Meson theory of nuclear force, Yukawa interaction, Charge independence and charge symmetry of nuclear forces.

Unit-III

Nuclear Structure (models): Single particle shell model and its successes, Semi-empirical formula of Weizsaker energy, β -activity of isobars, Liquid-drop model and Bohr-Wheeler theory of fission, Collective model of Bohr and Mottelson.

Unit-IV

Nuclear Interactions and Nuclear Reactions: Compound nucleus theory, Resonance reaction, Breit Wigner dispersion formula for $l=0$ neutrons, Weak interaction-phenomenon of β -decay; Fermi's theory; selection rules for β transition; parity non-conservation in β decay. Experimental demonstration.

Unit-V

Particle Physics: Fundamental interactions, Conservation laws, Discrete symmetries - parity; charge conjugation and time reversal; G parity and CPT theorem, Internal symmetries -Isospin formalism; SU2 and SU3 groups and their applications to multiplet mesons and baryons; Quark model -Gell Mann - Okubo mass formula for octet and decuplet hadrons - charm, bottom and top quarks, Gluons as mediators of strong interaction.

Books Suggested:

- Introductory nuclear Physics by Kenneth S. Krane, Wiley India Pvt. Ltd., 2008.
- Concepts of nuclear physics by Bernard L. Cohen, Tata Mcgraw Hill, 1998.
- Introduction to Elementary Particles by D. Griffith, John Wiley & Sons
- Introductory Nuclear Physics by S.S.M. Wong, PHI
- Theoretical Nuclear Physics by J.M. Blatt, & V.F. Weisskoff, John Wiley
- Introduction to Nuclear Physics by H.A. Enge, Addison Wesley
- Nuclear Physics by R.R. Roy, & B.P. Nigam, John Wiley
- Introductory Nuclear Theory by L.R.B Elton, Sir Isaac Pitman & Sons Ltd.



R.K.D.F. UNIVERSITY, RANCHI

M.Sc. PHYSICS

SEMESTER – III

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc. (Physics)	STATISTICAL PHYSICS	PPH302

UNIT - I

Specification of the state of the system, Macroscopic and Microscopic states, Phase space, Statistical ensemble, Postulate of equal a priori probability, Probability calculations, Behaviour of density of states, Liouville's theorem(Classical), Quasi-static processes.

Equilibrium conditions and constraints, Distribution of energy between systems in equilibrium, Approach to thermal equilibrium, Temperature, Heat reservoir, Sharpness of the probability distribution, Dependence of the density of states on the external parameters, Equilibrium between interacting systems.

UNIT - II

Micro-canonical ensemble, System in contact with heat reservoir, Canonical ensemble, Applications of canonical ensembles (Paramagnetism, Molecule in an ideal gas, Law of atmosphere), System with specified mean energy, Calculation of mean values and fluctuations in a canonical ensemble, Connection with thermodynamics, Grand- canonical ensemble, Physical interpretation of μ , Chemical potential in the equilibrium state, Mean values and fluctuations in grand canonical ensemble, Thermodynamic functions in terms of the Grand partition function.

UNIT - III

Classical partition functions and their properties, Calculations of thermodynamic quantities, Ideal monoatomic gas, Gibbs paradox, Equipartition theorem and its Simple applications. i) Mean kinetic energy of a molecule in a gas ii) Brownian motion iii) Harmonic Oscillator iv) Specific heat of solid. Maxwell velocity distribution , Related distributions and mean values.

UNIT -IV

Symmetry of wave functions, Quantum distribution functions, Boltzmann limit of Boson and Fermion gases, Evaluation of the partition function, Partition function for diatomic molecules, Equation of state for an ideal gas, The quantum mechanical paramagnetic susceptibility.

UNIT - V

Photon gas – i) Radiation pressure ii) Radiation density iii) Emissivity iv) Equilibrium number of photons in the cavity. Einstein derivation of Planck's law, Bose-Einstein Condensation, Specific heat, Photon gas – Einstein and Debye's model of solids. Fermi energy, Mean energy of fermions at absolute zero, Fermi energy as a function of temperature , Electronic specific heat, White – Dwarfs, Compressibility of Fermi gas, Pauli's paramagnetism, A relativistic degenerate electron gas

Reference Books :

1. Fundamentals of Statistical and Thermal Physics, - F.Reif, McGraw – Hill, International Edition (1985)
2. Fundamentals of Statistical Mechanics, B.B.Laud, New Age International Publication (2003)
3. Statistical Mechanics, R.K.Pathria, Butterworth Heinemann(2nd Edition)
4. Statistical Mechanics,K.Huang, John Wiley & Sons (2nd Edition)
5. Statistical Mechanics,Satya Prakash, Kedar Nath Ram Nath Publication (2008)



R.K.D.F. UNIVERSITY, RANCHI

M.Sc. PHYSICS

SEMESTER – III

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc (Physics)	CONDENSED MATTER PHYSICS -I	PPH304

Unit-I

X-ray Diffraction Theory: Coherent and incoherent scattering, Derivation of Laue equations and expression for structure factor, Data reduction.

Unit-II

Crystal Structure Determination: The phase problem in crystallography, Electron density as Fourier transform of structure factor and vice versa, Techniques to solve the phase problem – Fourier and Patterson methods, Heavy atom technique, The Single Isomorphous Replacement (SIR) and Multiple Isomorphous Replacement (MIR) techniques, Anomalous scattering technique, Direct methods.

Unit-III

Experimental Techniques: The Weissenberg and Precession methods, The Diffractometer, Area Detector and Image Plate. Fermi Surface: Construction of Fermi surface, Zone schemes, Electron, hole and open orbits, Cyclotron resonance. Determination of Fermi surface – Quantization of orbits in magnetic field; de Hass – van-Alfen effect; External orbits; Outline of other methods.

Unit-IV

Phonons: Harmonic crystals, Crystal potential; Harmonic and adiabatic approximations; Normal modes and phonons; Phonon spectrum by neutron scattering; Crystal momentum. Anharmonic crystals, Anharmonicity, Lattice thermal conductivity, Umklapp process Second sound.

Unit-V

Magnetism: Interaction of solids with magnetic fields, Magnetization density and susceptibility, Calculation of atomic susceptibility, Susceptibility of insulators (Larmor diamagnetism), Ground state of ions with partially filled shells (Hund's rule), van Vleck para magnetism, Curie laws for free ions and solids, Pauli para magnetism, Conduction electron diamagnetism, Exchange interaction, Ferromagnetic domains, Anisotropy energy, Thickness and energy of Bloch walls, Ising model, Bragg-Williams approximation, Solution of Ising problem for a linear chain.

Books Suggested:

- Philips, "An Introduction to Crystallography".
- Woolfson, M.M., "An Introduction to X-ray Crystallography".
- International Tables for X-ray Crystallography, Vol. I Verma.
- R. & Krishna, P., "Polymorphism and Polytypism".
- Kittel. C., "Solid-State Physics".
- Raghavan, V., "Material Science and Engineering".
- Ashcroft, N.W. and Mermin, N. D., "Solid-State Physics".
- Bunge, M.J., "Crystal Structure Analysis".
- Bunge, M.J., "X-ray Crystallography".
- Staut & Jenson, "A Practical Guide to X-ray Crystal Structure Determination".



R.K.D.F. UNIVERSITY, RANCHI

M.Sc. PHYSICS

PRACTICAL-III

GENERAL EXPERIMENTAL LAB

1. Frank Hertz Experiment.
2. Experiment with Hall apparatus.
3. Four-Probe set up for mapping the resistivity of large sample.
4. Measurement of magneto resistance of semiconductor sample.
5. Measurement of Susceptibility of paramagnetic solution by Quinke's tube method.
6. Study of the energy band gap and diffusion potential of p-n junction.
7. Study of Multi-vibrator.
8. Study of Characteristics of Semiconductor diodes: Si, Ge, Zener and LED.
9. Study of an Integrated Circuit Regulator.
10. Two Probe method for resistivity measurement of insulators at different temperatures



R.K.D.F. UNIVERSITY, RANCHI

M.Sc. PHYSICS

SCHEME Semester – IV

S.N.	Subject Code	Subject Name	Marks Distribution				
			Internal	External		Total	
			Max	Max	Min	Max	Min
1.	PPH401	Nanophysics and Nanomaterials	30	70	21	100	35
2.	PPH402	Condensed Matter Physics –II	30	70	21	100	35
3.	PPH404	Dissertation	30	70	21	100	35
<i>Practical</i>							
Total							
			Max		Min		
1.	PPH355	Practical-IV Lab	100		50		

Choice Based Credit System

S.N.	Subject Code	Subject Name	L	T	P	Credit
1.	PPH401	Nanophysics and Nonmaterial	4	1	0	5
2.	PPH402	Condensed Matter Physics –II	4	1	0	5
4.	PPH403	Dissertation	0	0	12	6
5.	PPH454	Practical-IV Lab	0	0	8	4
Total			8	2	20	20



R.K.D.F. UNIVERSITY, RANCHI

M.Sc. PHYSICS

Syllabus SEMESTER – IV

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.SC. (Physics)	NANO MATERIAL AND NANO PHYSICS	PPH401

UNIT -I

Low dimensional materials. Application in electronics, communication, medicine etc. Electron states in a potential well, spherically symmetric potential, Coulomb potential and periodic potential. Tunneling through a potential barrier. Excitons, biexcitons, dark excitons.

UNIT – II

Clusters. Fullerenes, semiconductor and metal clusters, cluster stability. Nanotubes. Electron states in nanoparticles, effective mass approximation, weak confinement, strong confinement, size dependent oscillator strength.

UNIT – III

Synthesis of nanomaterials (bottom up approach) by physical techniques. Introduction to vacuum techniques (pumps, gauges, materials). Physical vapour deposition, electron beam evaporation, sputter deposition, laser ablation, ion beam mixing, plasma deposition.

UNIT – IV

(Synthesis of nanomaterials by chemical, biological and hybrid routes). Concepts of colloids, LaMer diagram, L.B films, Micellar route, self assembly, biosynthesis, electrophoresis, immobilization in glass, zeolites, polymers.

UNIT – V

Analysis Techniques. UV-VIS-IR spectroscopy, Luminescence techniques, X-ray, electron and neutron Diffraction, Small Angle X-ray and Neutron Scattering, photon correlation spectroscopy, Extended X-ray , Absorption Fine Structure (EXAFS), X-ray Photoelectron Spectroscopy, Auger Electron Spectroscopy.

Reference Books:

1. *Physics of Low Dimensional Structures*, J. H. Davis, (Cambridge Press), 1998.
2. *Semiconductor Quantum Dots*, L. Banjaj and S. W. Koch.
3. *Low Dimensional Semiconductors*, M. J. Kelly, Clarendon, 1955.
4. *Characterization of Materials*, J. B. Wachtman and Z. H. Kalman, Butterworth-Heinmann, USA, 1993.
5. *Experimental Physics*, Modern Methods, R. A. Dunlop.
6. *Instrumental Methods of Analysis*, H. H. Willard, L. L. Merritt, J. A. Dean and F. A. Settle, (CBS Pub.), 1986.



R.K.D.F. UNIVERSITY, RANCHI

M.Sc. PHYSICS

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc. (Physics)	CONDENSED MATTER PHYSICS -II	PPH402

Unit-I

Phase Transformation and Diagrams: phase rule, single component system, Binary phase system, lever rule, Nucleation and growth, Nucleation kinetics, Growth and overall transformation kinetics and applications to steel and glass

Unit-II

ESR: basic theory, relaxation mechanism, Effect of spin-orbit coupling and crystal fields on g values, Fine and hyperfine structures, Ferromagnetic resonance (FMR), General features of FMR, Shape effect in FMR, Antiferromagnetic resonance.

Unit-III

NMR: Basic theory, Spin lattice relaxation, Bloch equation and their steady state solutions, General features of NMR spectra, Chemical shifts, Fine structure due to spin-spin coupling, Application to molecular structure and bondings.

Unit-IV

Superconductivity: BCS theory of superconductivity, Cooper pairs, superconducting ground state, Flux quantization in superconducting ring, Quasi-particles and energy gaps, Temperature dependence of energy gaps, London equation, Coherence length, Persistent current, Single particle tunneling, Josephson tunneling, Josephson effects (AC and DC), Microscopic quantum interference, Qualitative idea of high temperature superconductors, Critical fields and moments.

Unit-V

Thin Films: Deposition techniques, thermal, electron and sputtering methods, metallic semiconductor and insulator thin films and their electrical, electronic and optical properties. Magnetic superconducting thin films and applications.

Unit-VI

Dielectrics: Structure of dielectrics, Polarization mechanism, Effect of temperature and frequency. Effect of conduction (ionic and electronic) in dielectrics, Dielectric losses and breakdown, Electrets and MIM.

Books Suggested:

- Crystallography – Philips
- Solid State chemistry-Garner (Butterworth; London)
- Solid State Chemistry -D.K.Chakraborty (New Age int Publication)
- Solid State Chemistry- N. BHannay (Prentice Hall, New Jersay)
- Physical Chemistry- Waller J. Moore
- Principles of polymer chemistry Cornell , P. J. Flory (Univ. Press)
- Handbook of Conducting Polymers Vol I• & II" T A. Skolhia.



R.K.D.F. UNIVERSITY, RANCHI

M.Sc. PHYSICS

PRACTICAL

LIST OF EXPERIMENTS FOR M.Sc. FINAL

LIST OF EXPERIMENTS (any eighteen) :

1. To determine half-life of a radio isotope using GM counter.
2. To study absorption of particles and determine range using at least two sources.
3. To study characteristics of a G M counter and to study statistical nature of radioactive decay.
4. To study spectrum of γ -particles using Gamma ray-spectrometer.
5. To calibrate a scintillation spectrometer and determine energy of γ -rays from an unknown source.
6. (a) To study variation of energy resolution for a Nai (T) detector.
(b) To determine attenuation coefficient (μ) for rays from a given source.
7. To study Compton scattering of γ -rays and verify the energy shift formula
8. To study temperature variation of resistivity of a semi-conductor and to obtain band gap using four probe method.
9. To study hall effect and to determine hall coefficient.
10. To study the variation of rigidity of a given specimen as a function of the temperature.
11. To study the dynamics of a lattice using electrical analog.
12. To study ESR and determine g -factor for a given spectrum.
13. To determine ultrasonic velocity and to obtain compressibility for a given liquid.
14. Study the characteristics of a gyrotron and calculate the mode number, E.T.S. and transit time.,
15. Study the simulated L.C.R. transmission line (audio frequency) and to find out the value for Z_0 experimentally from the graph.
16. Study the radiation pattern of a given Pyramidal horn by plotting it on a Polar graph paper. Find the Half power beam width and calculate its gain.
17. Find the dielectric constant of a given solid (Teflon) for three different lengths by using slotted section.
18. Find the dielectric constant of a given liquid (organic) using slotted section of K-band.
19. Verification of Bragg's law using microwaves.
20. Determination of Dielectric Constant of a liquid by lecher wire.
21. Study of a Heat Capacity of Solids.
22. Study of lattice dispersion.